Machine Learning

Maximum A-Posteriori & Maximum Likelihood Estimation

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Recall — Bayesian Learning

$$P(X|\mathbf{e}) = \sum_i P(X|\mathbf{e}, h_i)P(h_i|\mathbf{e})$$

$$= \sum_i P(X|h_i)P(h_i|\mathbf{e})$$

predictions

predictions of individual hypothesis

weight: posterior/belief

weighted average of predictions of individual hypotheses

- **Bayesian learning properties**
  ++ Optimal (given prior, no prediction is correct more often than the Bayesian prediction)
  ++ No overfitting (since all hypotheses are considered and weighted)

- **Limitation**
  - - When hypothesis space is large, sum over hypotheses may be **intractable**

- **Solution**
  – "Approximate" Bayesian learning

Bayesian Learning Approximations

- Maximum A-Posteriori (MAP)
- Maximum Likelihood (ML)

Maximum a posteriori (MAP)

- The first approximation is MAP
- Key idea
  - Limitation of Bayesian Learning: considering too many hypotheses
  - Solution: Instead of considering the full posterior distribution (all hypotheses)
    - Choose the hypothesis with highest probability in the posterior
    - Then make predictions based on chosen hypothesis

Maximum a posteriori (MAP)

- **MAP:**
  Make predictions based on the <u>most probable hypothesis</u> h_{MAP}

most probable hypothesis
max a-posteriori hypothesis

$$h_{MAP} = argmax_{h_i} P(h_i \mid \mathbf{e})$$

approximate prediction

$$P(X \mid \mathbf{e}) \approx P(X \mid h_{MAP})$$

- **Bayesian learning:**
  Make predictions based on <u>all hypotheses weighted by their (posterior) probability</u>

MAP properties

- **Computation / Accuracy tradeoff**
  MAP relies only on <u>one hypothesis</u> h_{MAP}
  - **++** Simpler computation
  - **- -** MAP predictions are less accurate than Bayesian predictions
    [chosen hypothesis may not be the best. Example: fits data too well: overfit]

- **Overfitting**
  ++ It is possible to **<u>control overfitting</u>** using the **<u>prior</u>**
  1. Assign **<u>complex</u>** hypotheses **<u>lower prior</u>** probability
     [avoid complex hypotheses that can easily overfit & capture noise in the data]
  2. Assign **simpler** hypotheses **higher prior** probability
     [less likely to overfit]

  - - <u>May not be able to avoid overfitting completely</u>
  There is still **a chance of overfitting** since only one hypothesis is considered
  If that hypothesis still captures some noise in the data, there will be overfitting

MAP properties

- **Convergence**
  MAP & Bayesian predictions converge when sufficient data is available

- **Tractability**
  ++ Computation: Considering a single hypothesis h_{MAP} is a big win
  - - Finding h_{MAP} is an optimization problem:
  Find hypothesis that maximizes the posterior:
  - - May be intractable [optimization may be difficult]

$$h_{MAP} = argmax_h P(h \mid \mathbf{e})$$

!!!! MAP does not solve the tractability problem — MAP just changes it !!!!

Maximum Likelihood (ML)

- The second approximation is ML
- **Key Idea**
  - Instead of finding hypothesis with highest probability a-posteriori (in the posterior)
    - Find the hypothesis that best fits data:
    - The hypothesis with the highest likelihood of generating data (evidence e)

Maximum Likelihood (ML)

- ML simplifies MAP by assuming uniform prior

$$P(h_i) = P(h_j) \forall_{i,j}$$

uniform prior assumption

$$h_{MAP} = argmax_h P(h \mid \mathbf{e})$$

posterior

$$h_{MAP} = argmax_h P(h) P(\mathbf{e} \mid h)$$

prior likelihood

$$h_{ML} = argmax_h P(\mathbf{e} \mid h)$$

likelihood

- Make prediction based on h_{ML} only:

$$P(X \mid \mathbf{e}) \approx P(X \mid h_{ML})$$

prediction

- ML: an approximation because we eliminate the prior
- ML: a simplification because there is no need to set the prior
- Prior: distribution over all hypotheses
  complex to encode if there are too many hypotheses

Maximum Likelihood (ML)

Key Intuition

- There is only a finite amount of probability mass (sum-to-one constraint)
- ML tries to allocate **as much** probability mass **as possible** to the things we have **observed**
- At the **expense** of the things we have not **observed**

ML properties

- **Computation / Accuracy tradeoff**
  ML ignores prior info & relies only on one hypothesis h_{ML}
  + + Simpler computation
  - - ML prediction less accurate than Bayesian & MAP predictions

- **Convergence**
  — ML, MAP & Bayesian predictions converge when sufficient data is available

- **Overfitting**
  — Subject to overfitting
  — Pick hypothesis that best fits data [risk: fit everything including noise]
  — No prior to penalize complex hypotheses that have the risk of exploiting statistically insignificant data patterns

$$h_{ML} = argmax_h \sum_n logP(e_n|h)$$
likelihood

- **Tractability**
  — Still have an optimization problem
  — Could still be intractable
  — Finding h_{ML} is often **easier** than h_{MAP}
  — Many ML algorithms maximize likelihood (solve this optimization)

Infinite Data?

- Bayesian learning, MAP & ML **converge** to same prediction given infinite data

- Bayesian Learning
  - Start with a prior that captures current uncertainty
  - More data: reduces uncertainty
    leading to more confidence in a single hypothesis being the right one

- In the limit, with infinite data, learning converges to the same hypothesis **H**:
  - Bayesian Learning: all mass of posterior distribution centers on one hypothesis **H** which explains the data well *(or a few equally good hypotheses)*
  - **H** has highest probability in the posterior (MAP)
  - **H** is the most consistent with data (ML)

- Since they are all equivalent, then given sufficient data, use ML (simplest)

How much data is sufficient?

- Different problems require different amounts of data
  - A large hypothesis space requires more data to find the best hypothesis
  - A Small hypothesis space less data may be okay
- Answering this question requires a full course on learning theory!

Is this a realizable problem?

- Is there a true underlying function that exists as a hypothesis within the hypothesis space considered?

  - We do not make this assumption

- Recall the candy example:
  — We set 5 hypotheses for 5 types of bags
  — If the store sells other types, then we do not have a realizable solution

- The three approaches will converge to the best hypothesis within the hypothesis space considered

Principles of MAP & ML

MAP Estimation
Choose hypothesis to **maximize posterior of hypothesis** given data

$$h_{MAP} = argmax_h P(h \mid \mathbf{e})$$

ML Estimation
Choose hypothesis to **maximize likelihood** of data

$$h_{ML} = argmax_h P(\mathbf{e} \mid h)$$

Principles of MAP & ML

MAP Estimation
Choose hypothesis to **maximize posterior of hypothesis** given data

$$h_{MAP} = argmax_h P(h)P(\mathbf{e}\,|\,h)$$

Prior **Likelihood**

ML Estimation
Choose hypothesis to **maximize likelihood** of data

$$h_{ML} = argmax_h P(\mathbf{e}\,|\,h)$$

Bayesian, MAP & ML

	Bayesian	MAP	ML
Learning	$P(H \mid \mathbf{e}) = k\ P(\mathbf{e} \mid H)P(H)$	$h_{MAP} = argmax_{h_i} P(h_i \mid \mathbf{e})$	$h_{ML} = argmax_h P(\mathbf{e} \mid h)$
Inference	$P(X \mid \mathbf{e}) = \sum_i P(X \mid h_i)P(h_i \mid \mathbf{e})$	$P(X \mid \mathbf{e}) \approx P(X \mid h_{MAP})$	$P(X \mid \mathbf{e}) \approx P(X \mid h_{ML})$
Accuracy	**Optimal**	**Less accurate** than Bayesian	**Less accurate** than Bayesian & MAP
Overfitting	**No overfitting**	**Controlled overfitting**	**Overfitting**
Choosing *h*	All hypotheses are used	Solve optimization to find $\boldsymbol{h_{MAP}}$ (**Intractable**)	Solve optimization to find $\boldsymbol{h_{ML}}$ **(Intractable)** Easier than $\boldsymbol{h_{MAP}}$ **(Less intractable)**
Making predictions	**Intractable** to compute prediction	Prediction computed using one hypothesis	Prediction computed using one hypothesis

Summary

- Machine Learning involves **function approximation**
- The principle of **maximum likelihood estimation** provides **an alternate view of learning**
  - **Probability distributions** can be used to **model real data** that occurs in the world